JAMES RUSE AGRICULTURAL HIGH SCHOOL

2022

YEAR 10 SCIENCE

THEORY

YEARLY EXAMINATION

Total: 80 marks

- **SECTION A – Multiple Choice Questions** 20 marks
- **SECTION B – Genetics, Biotechnology and Evolution** 20 marks
- **SECTION C – Using Chemistry** 20 marks
- **SECTION D – Electromagnetism and the Universe** 20 marks

General Instructions

- Reading time - 5 minutes
- Working time - 90 minutes
- Board-approved calculators may be used
- Write using black pen
- Draw diagrams using pencil
- A Periodic Table is provided
- Write your name and class in the space provided on the top of each section.
- Please tear off the back page of this booklet.

Section A: 20 Multiple Choice Questions **20 marks**

1. The colour of a star gives us information about which of the following?

   A. Distance from Earth
   B. Luminosity
   C. Chemical composition
   D. Temperature

2. The different types of finches developing on the Galapagos Islands is an example of which of the following?

   A. Speciation
   B. Natural selection
   C. Mutation
   D. Selective pressure

3. Which of the following equipment would be used to observe electromagnetic induction?

   A. Magnet, solenoid, ammeter
   B. Magnet, solenoid, voltmeter
   C. Magnet, transformer, voltmeter
   D. Magnet, transformer, ammeter

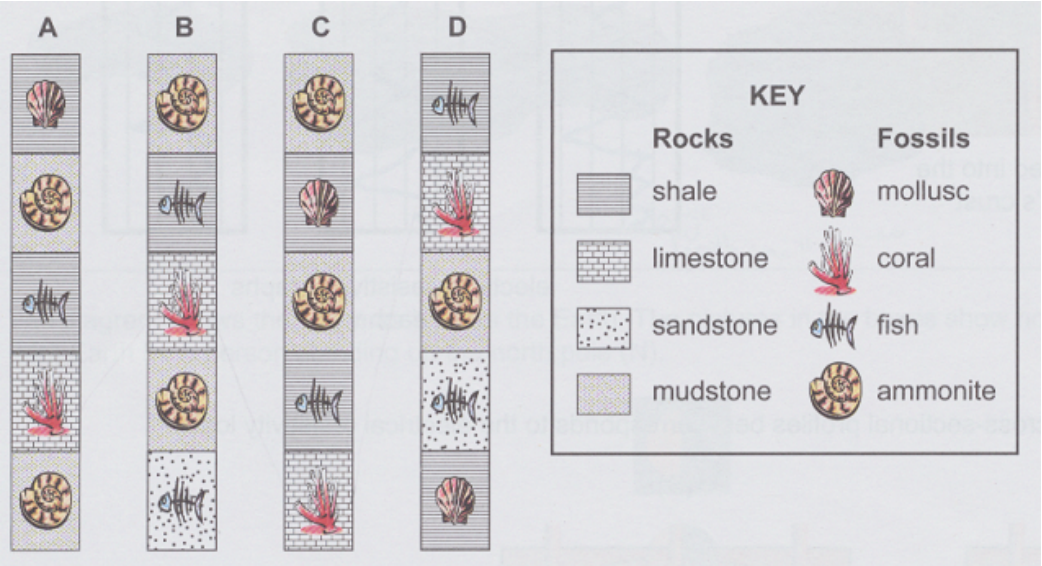
4. Over the last 50 years chicken breeders have changed the characteristics of chickens grown for meat via artificial selection. Which one of the statements below best describes the process through which this change has been brought about?

   A. Farmers have taken genes from a turkey, inserted them into a chicken embryo and then cloned the resulting offspring
   B. Farmers have fed chickens on very high protein food, causing them to get bigger and bigger with each generation.
   C. Farmers have only allowed the offspring in each generation with the characteristics they want, to breed.
   D. Farmers have given chickens growth hormones, and over a number of generations they have become bigger and bigger.

5. Nebulae are gigantic astronomical objects observed in space. What are they made of?

   A. hydrogen and helium
   B. dust and gas
   C. planets and stars
   D. stars

Refer to the rock samples to answer Questions 6 and 7.



6. Which fossil is the index fossil in this area?

A. Mollusc
B. Coral
C. Fish
D. Ammonite

7. Which of the columns contains the youngest rock?

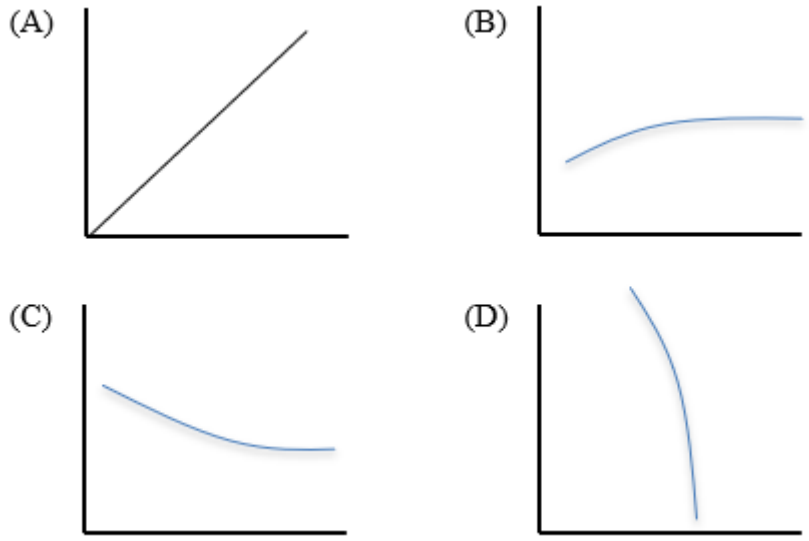
A. A
B. B
C. C
D. D

Use the information below to answer Questions 8 and 9.

The table shows information about the rate of photosynthesis in *Elodea*, a water weed plant under different conditions. Rate of photosynthesis is determined by the amount of oxygen produced.

Rate of Photosynthesis in Elodea					
Lamp distance (cm)	15	30	45	60	75
Oxygen gas produced (mL/hr)	7.25	6.25	5.0	4.5	4.5

8. If the axes were labelled, which graph best reflects these results?



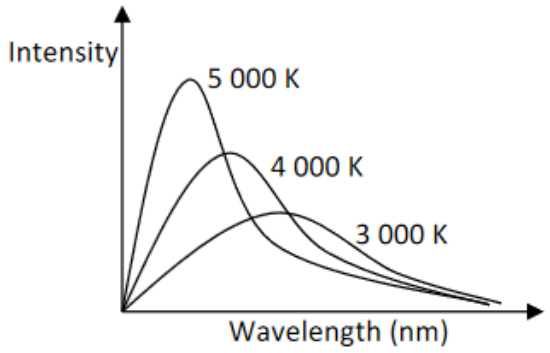
9. What would be the most appropriate conclusion for these results?

A. The rate of photosynthesis for all plants is highest when 15 cm from a lamp.
B. The rate of photosynthesis decreases as distance from the light source increases.
C. The rate of photosynthesis increases as distance from the light source increases.
D. The rate of photosynthesis is not affected by the distance of the lamp from the plant, but by the temperature of the water.

10. Observatories with telescopes are often situated on isolated mountains. What is the main reason for this?

A. They experience less interference.
B. They work better at low temperatures.
C. The land is cheaper.
D. Their altitude places them closer to the stars.

11. The graph below shows the intensity vs wavelength of a few stars.



From the graph it can be concluded that:

A. As intensity increases, temperature decreases and peak wavelength increases.
B. As intensity increases, temperature increases and peak wavelength decreases.
C. As intensity decreases, temperature decreases and peak wavelength decreases.
D. As intensity decreases, temperature increases and peak wavelength increases.

12. Why can a mutation be an advantage for populations of living organisms?

A. It makes the organism more resistant to diseases.
B. It produces cells that can divide and multiply faster.
C. It can lead to the growth of new organs in organisms.
D. It produces variations in the characteristics of the organisms.

13. Colour blindness is an example of sex-linked inheritance. A non-colour blind woman carrying an allele for colour blindness marries a man who is colour blind. What possible offsprings will they have?

A. All of the daughters but none of the sons will be colourblind.
B. Half will be colour blind, regardless of sex.
C. All will will be colour blind, regardless of sex.
D. Half of the daughters and all of the sons will be colour blind.

14. Which of the following does not affect the reaction rate?

A. Concentration of a solution
B. Addition of a catalyst
C. Amount of reactant
D. Surface area of a solid

Use the following information to answer Questions 14 and 15.

An experiment was performed to compare several brands of indigestion tablets, to see which was best at neutralising stomach acids. The active ingredient in each tablet was calcium carbonate.

Hydrochloric acid was added to each tablet, one drop at a time, until no further reaction occurred. The number of drops of acid needed to neutralise each tablet was counted.

15. What was the independent variable in this experiment?

    A. The concentration of the hydrochloric acid.
    B. The number of drops of hydrochloric acid.
    C. The calcium carbonate.
    D. The brand of indigestion tablet.

16. Which would be the best indigestion tablet?

    A. The one which reacted with the acid fastest.
    B. The one which reacted with the acid slowest.
    C. The one which needed the most drops of acid.
    D. The one which needed the least drops of acid.

17. For the following chemical equation:

    $Na + H_2O \rightarrow NaOH + H_2$

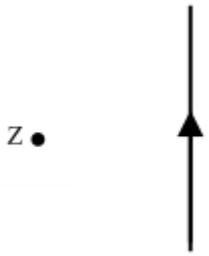
    Identify the correct balanced equation.

    A. $2\ Na + H_2O \rightarrow (NaOH)_2 + H_2$
    B. $Na_2 + H_4O_2 \rightarrow Na_2O_2H_2 + H_2$
    C. $2\ Na + 2\ H_2O \rightarrow 2\ NaOH + H_2$
    D. $Na_2 + 2\ H_2O \rightarrow 2\ NaOH + H_2$

18. Which of the following is not an example of a plastic polymer?

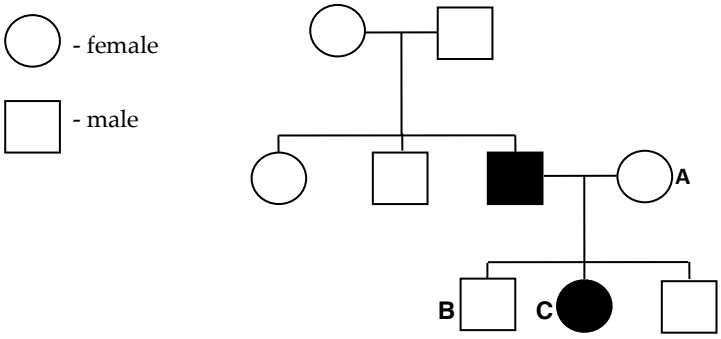
    A. Polyfluorocarbon.
    B. Polyethene
    C. Polyvinyl chloride
    D. Polystyrene

19. Identify the correct direction of the magnetic field at point Z near the current-carrying wire shown.



A. Up the page
B. Down the page
C. Into the page
D. Out of the page

20. The pedigree shows the inheritance of a particular genetic trait. The black squares and circles indicate that the trait is expressed.



Which of the following statements is correct?

A. Individual A and B are heterozygous for the recessive trait and C is homozygous for the recessive.
B. Individual A is homozygous for the dominant trait and B and C are homozygous for the recessive trait.
C. Individual C is homozygous for the dominant trait and A and B are heterozygous for the recessive trait.
D. Individual A is homozygous for the dominant trait, B is heterozygous for the dominant trait and C is homozygous for the recessive.

-End of Section A-

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen																	2 He 4.003 Helium
3 Li 6.941 Lithium	4 Be 9.012 Beryllium											5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium											13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanoids	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89–103 Actinoids	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson

KEY

Atomic Number: 79
Symbol: Au
Standard Atomic Weight: 197.0
Name: Gold

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium

Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.
Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

SECTION A

ANSWER BOOKLET

Name	
Class	
Mark	

Section A Answer grid for multiple choice questions 1 – 20. <u>Please tear off this page.</u>

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O
6.	A O	B O	C O	D O
7.	A O	B O	C O	D O
8.	A O	B O	C O	D O
9.	A O	B O	C O	D O
10	A O	B O	C O	D O
11.	A O	B O	C O	D O
12.	A O	B O	C O	D O
13.	A O	B O	C O	D O
14.	A O	B O	C O	D O
15.	A O	B O	C O	D O
16.	A O	B O	C O	D O
17.	A O	B O	C O	D O
18.	A O	B O	C O	D O
19.	A O	B O	C O	D O
20.	A O	B O	C O	D O

JAMES RUSE AGRICULTURAL HIGH SCHOOL

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Mark	**/20**

2022

YEAR 10 SCIENCE

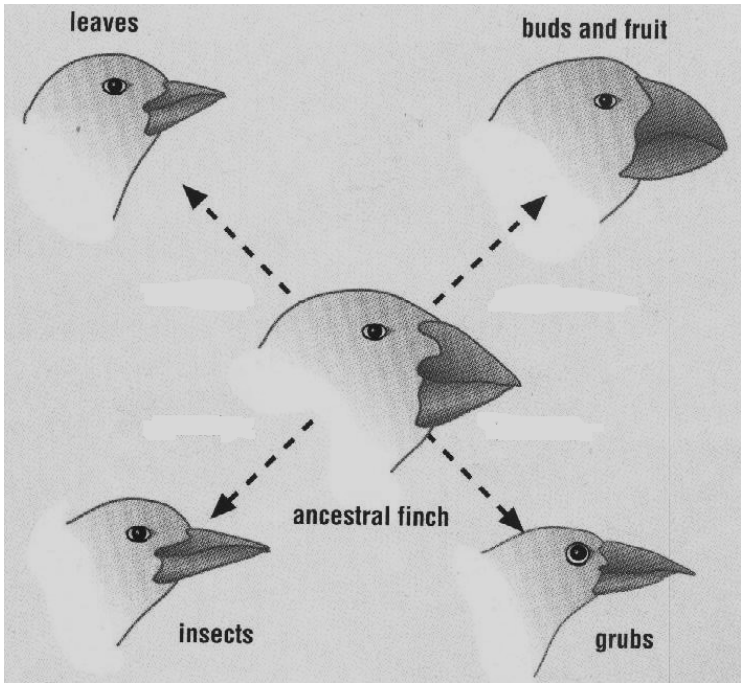
YEARLY EXAMINATION

SECTION B

- **SECTION B – Genetics, Biotechnology and Evolution** 20 marks

Question 1 (3 marks)

The diagram below displays various finches found on the Galapagos Islands. All types of finches are thought to have descended from one type of finch that came from the mainland of South America.



Explain how natural selection accounts for the variety of finches that now occur on the Galapagos Islands.

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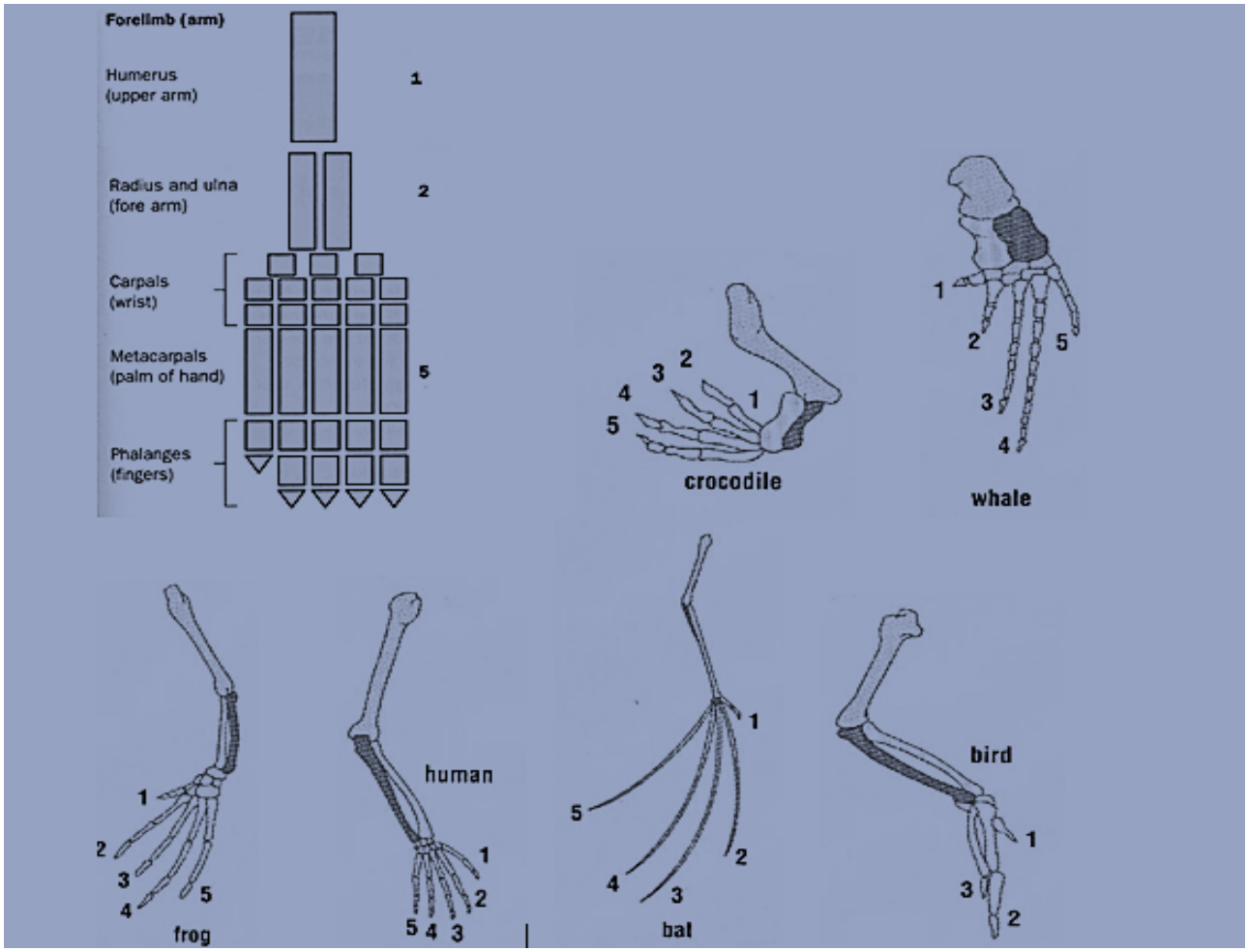
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Question 2 (3 marks)

The diagram below shows the pentadactyl limbs of various mammals and a basic pentadactyl limb layout. Outline how these structures provide evidence for evolution.



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Question 3 (4 marks)

Explain the role of variations and mutations in a gene pool for the success of a species.

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Question 4 (4 marks)

Describe an example of a biotechnology which you have studied and discuss the reasons in support and opposition of using that technology.

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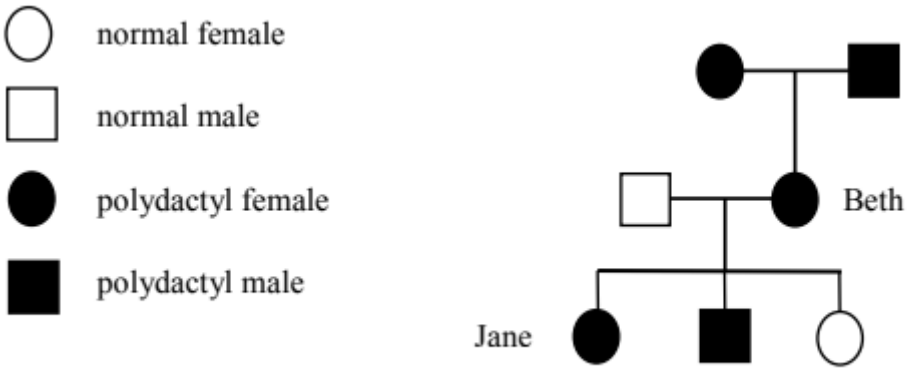
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Question 5 (6 marks)

The pedigree below shows the inheritance of a type of polydactyly (the possession of extra fingers). This type is caused by the inheritance of a dominant allele. The condition is not sex-linked.



(a) Is it possible to tell from this pedigree whether Beth is homozygous or heterozygous for this condition? Justify your answer. **2**

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(b) Jane marries a man who doesn't have the trait. List the possible genotypes of their children. **2**

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(c) What proportion of their children would you expect to have the condition? Show working. **2**

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-End of Section B-

JAMES RUSE AGRICULTURAL HIGH SCHOOL

Name	
Class	**10A 10C 10K 10L 10T**
Mark	**/20**

2022

YEAR 10 SCIENCE

YEARLY EXAMINATION

SECTION C

- **SECTION C – Using Chemistry** 20 marks

Question 1 (3 marks)

Hydrochloric acid was added to identical masses of sodium carbonate crystals in 4 separate test tubes. The test tubes had equipment attached to measure the volume of gas produced.

The results for each test tube varied, producing 38, 35, 25, and 8mL of gas respectively after 3 minutes.

(a) Suggest one possible variable that may have caused the acid to react at different rates with the four samples of sodium carbonate. **1**

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(b) Using your response to (a), suggest TWO ways that variables should have been controlled in this experiment to make it valid. **2**

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Question 2 (7 marks)

A student was asked to investigate the effect of temperature on the time taken for a piece of magnesium to dissolve in hydrochloric acid.

(a) She used a test tube, strips of magnesium, and 1M hydrochloric acid. List THREE other pieces of equipment she would need to use. **1**

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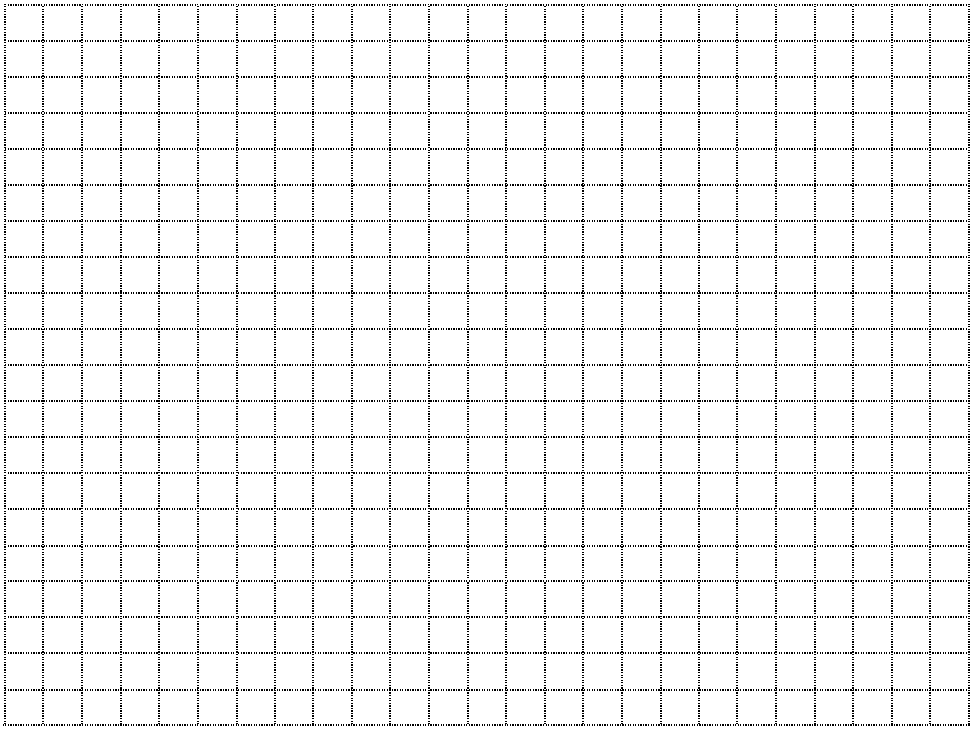
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(b) She used 5 different temperatures and timed 3 trials for each temperature. An extract from her notebook is shown below.

5°C	11°C	24°C	39°C	62°C
180 secs	158 secs	98 secs	50 secs	11 secs
171 secs	140 secs	102 secs	38 secs	9 secs
199 secs	149 secs	133 secs	35 secs	13 secs

Present these results in a table and calculate the average reaction time for each temperature. **3**

(c) Graph the average results on the grid below **3**



Question 3 (4 marks)

Draw a labelled diagram of a galvanic cell with a voltmeter, zinc and copper electrodes, a salt bridge, and zinc sulfate and copper sulfate as the electrolytes. Also label the anode, cathode, salt bridge and direction of electron flow.

The half equations for the cell are: $Zn \rightarrow Zn^{2+} + 2e^-$ and $Cu^{2+} + 2e^- \rightarrow Cu$.

Question 4 (2 marks)

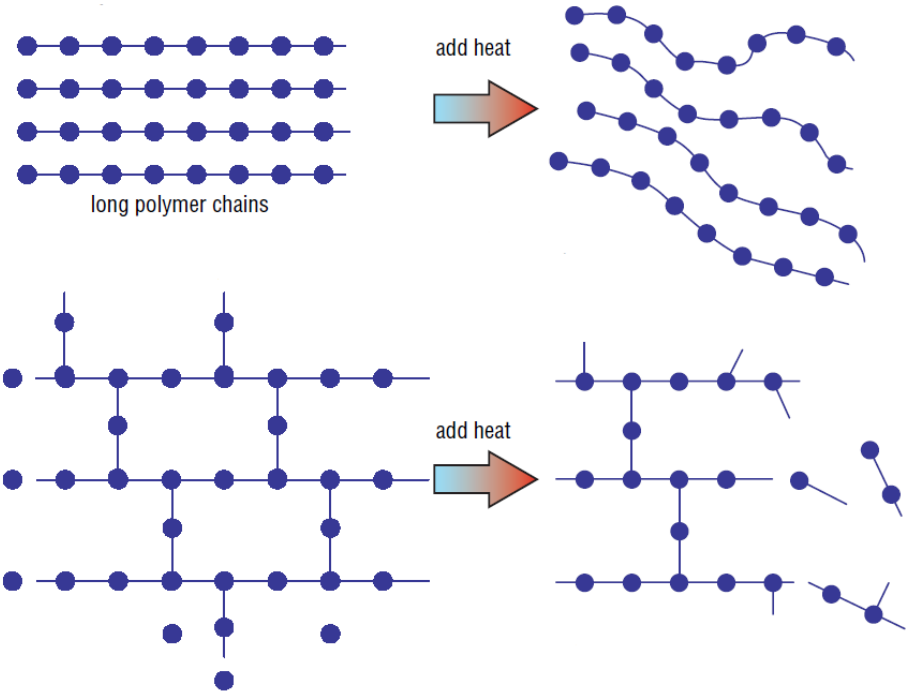
Solutions of aluminium bromide and potassium sulfate are mixed to form potassium bromide and aluminium sulfate. No precipitate is formed.

Write a balanced chemical equation with states for the above reaction. **2**

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Question 5 (4 marks)

The following diagram shows the structures of 2 types of plastic polymers and what happens when heat is applied to them.



Identify the 2 types of plastics and explain what happens to each one when burned.

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-End of Section C-

JAMES RUSE AGRICULTURAL HIGH SCHOOL

Name	
Class	**10A 10C 10K 10L 10T**
Mark	**/20**

2022

YEAR 10 SCIENCE

YEARLY EXAMINATION

SECTION D

- **SECTION D – Electromagnetism and the Universe** 20 marks

Question 1 (6 marks)

(a) Explain the difference between AC and DC electricity, giving examples of where each one is used. **2**

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(b) Electricity is supplied in Australian households at 240V. However, electric power is transported over large distances at high voltages (e.g. 66kV) rather than at 240V. Explain why. **1**

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(c) Explain the functions of a transformer. **2**

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(d) Name a device used at home or in a school that uses an electromagnet. **1**

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Question 2 (8 marks)

(a) The Sun is a relatively small star. Sort the following stages of the Sun's life into the correct order. **2**

Stages in the life cycle of our Sun: Nebula, white dwarf, red giant and protostar.

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(b) The two brightest stars of the Southern Cross are Alpha Crucis and Beta Crucis. Jane told John that if they are the same brightness that means they are the same distance away from Earth.

Is Jane correct? Explain your answer. **2**

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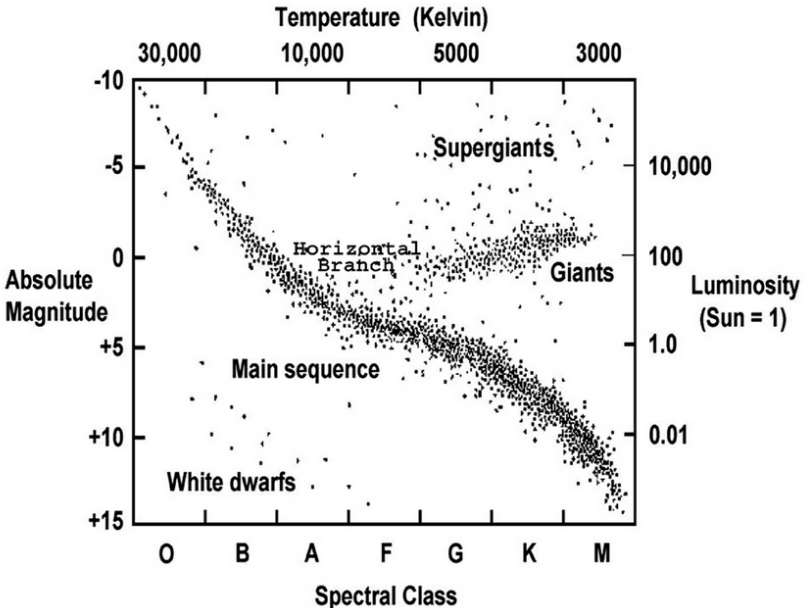
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(c) Identify the name of the 'explosion' that occurs when a large star dies, and the 2 possible outcomes of such an explosion. **2**

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(d) The graph below plots stars according to their temperature, luminosity, magnitude and spectral class.



Use the graph to compare the properties of a white dwarf and a giant star. **2**

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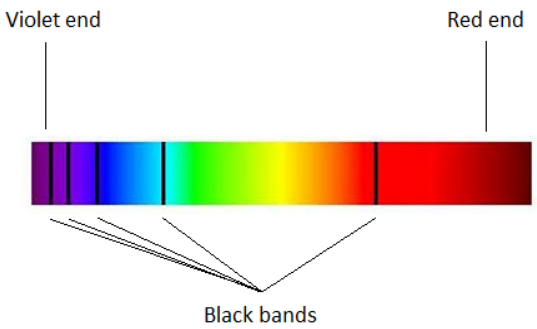
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Question 3 (6 marks)

(a) Identify what the following image is an example of. **1**



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(b) Name a technology, besides the optical telescope and spacecraft, that has enhanced our understanding of the universe. **1**

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(b) Outline the Big Bang theory and explain the evidence in support of it. **4**

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-End of Section D-